# Explainable AI Framework for Employee Attrition Prediction





## Project Team

Mahith 2420090095

Ch. Charan Teja 2420090022

G. Nishanth 2420030679

Moksha 2400090182

Course Coordinator :
Dr. K. Swanthana

Team :16

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# Problem Statement

◆ Employee attrition is a major challenge for organizations because it increases recruitment costs, reduces productivity and creates difficulties in retaining skilled employees.

◆ Traditional machine learning systems generally predict whether an employee is likely to leave, but often provide limited transparency about the reason behind the prediction.

◆ HR managers need a system that not only identifies employees at risk, but also explains the factors influencing that risk.

◆ Our project proposes an Explainable AI framework to make attrition predictions transparent and actionable.

# Objectives

- To Develop a Framework, utilizing machine learning model to predict whether an employee is likely to leave the organization.
- **To improve** model generalizability by evaluating the framework using appropriate datasets and validation approaches.
- **To integrate** Explainable AI techniques to provide transparent explanations for individual employee attrition predictions.
- **To identify** the key factors contributing to employee attrition and provide actionable insights for reducing employee turnover.
- **To enhance** prediction reliability by addressing class imbalance and evaluating model performance using suitable metrics.

# Literature Survey

| Research Paper | Techniques Used | Purpose | Limitation |
|---|---|---|---|
| **IBM Employee Attrition Analysis** — Yang & Islam (2020) (arXiv) | Correlation Matrix, Random Forest, K-Means Clustering, Logistic Regression | To identify the major factors influencing employee attrition and analyze why employees leave an organization. | Uses a single IBM dataset, so the findings may have limited generalizability to other organizations. |
| **Predict Employee Attrition Using Multiple Machine Learning Techniques** — Ghanem & Al-Alawi (2024) (DOI) | Random Forest, XGBoost, Decision Tree, Bernoulli Naive Bayes, KNN, SVM, Logistic Regression, Oversampling, GridSearchCV | To compare multiple ML algorithms and identify the most effective model for employee attrition prediction. | Dataset has class imbalance and the results may not generalize well to other organizations; more datasets are needed. |
| **Employee Attrition Prediction using Machine Learning: A Data-Driven Approach for Workforce Analytics** (2025) (DOI) | Logistic Regression, Gradient Boosting, Random Forest, SVM, Gaussian Naive Bayes, Feature Engineering | To predict employee attrition and identify important employee characteristics that can help HR managers develop retention strategies. | The study is primarily focused on prediction and model comparison; broader real-world validation is still needed. |
| **A Stacking Ensemble Framework for Predicting Employee Turnover: Explainable AI with SHAP** — Rahman et al. (2025/2026 publication record) (EurekaMag) | Machine Learning, Ensemble Learning, Deep Learning, Stacking, SHAP | To predict employee turnover using ensemble approaches and explain the model's predictions using SHAP. | Relies on the IBM HR Analytics dataset, which limits generalizability to different organizations and workforce environments. |
| **Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics** (2026) (DOI) | Machine Learning, SHAP, Feature Selection, Data Balancing, Hyperparameter Optimization | To combine accurate attrition prediction with explainable AI so HR teams can understand the factors behind predictions. | Synthetic datasets such as IBM HR Analytics may not represent real-world organizations; SHAP may not fully capture complex feature interactions. |

| Research Paper | Techniques Used | Purpose | Limitation |
|---|---|---|---|
| **Employee Attrition Prediction Using Bayesian Optimized Stacked Ensemble Learning and Explainable AI** — Konar et al., 2025 | Stacking, XGBoost, Random Forest, Bayesian Optimization, SHAP, LIME | To improve attrition prediction and explain why employees are predicted to leave. | Mainly evaluated on benchmark data; broader organizational validation is still needed. (Springer) |
| **Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics** — Al-Ali et al., 2026 | Adaptive Boosting, Histogram Gradient Boosting, SHAP, feature selection, data balancing | To combine accurate attrition prediction with explainability, fairness and actionable HR insights. | Existing HR datasets may not fully represent different real-world organizations. (PubMed Central (PMC)) |
| **Explainable AI for Employee Turnover Prediction: A SHAP-Based Intelligent Analytics Approach** — Haojie Jia, 2026 | XGBoost, Random Forest, LightGBM, Logistic Regression, SHAP, SMOTE, fairness audit | To provide global, local and interaction-based explanations for employee turnover predictions. | Feature importance and subgroup effects can vary between datasets, requiring validation for each deployment. (Nature) |
| **Predicting Employee Attrition: XAI-Powered Models for Managerial Decision-Making** — 2025 | Transformer, GAN-based oversampling, SHAP | To handle class imbalance and provide transparent attrition predictions for HR managers. | Authors identify the need for validation across more industries and lightweight real-time implementation. (MDPI) |
| **Employee Attrition Prediction: An Explanatory and Statistically Robust Ensemble Learning Model** — 2026 | SHAP, LIME, Optuna, Random Oversampling, Ensemble Learning | To combine accurate prediction with global and local explanations and identify important retention factors. | Uses benchmark datasets, so real-world organizational validation remains important. (MDPI) |

# Research Gap Identification

- **Limited Generalizability:** Most studies rely heavily on the IBM HR Analytics dataset, making it difficult to confirm whether the models perform effectively across different organizations and workforce environments.

- **Limited Explainability:** Several existing approaches focus primarily on prediction accuracy and model comparison, without providing sufficient explanations for individual employee predictions.

- **Lack of Actionable Insights:** Existing models mainly identify employees who are likely to leave but provide limited guidance on **which factors should be addressed to reduce attrition**.

- **Class Imbalance Challenge:** Employee attrition datasets often contain significantly fewer employees who leave compared with those who stay, which can affect prediction performance.

- **Limited Real-World Validation:** Many approaches are evaluated on benchmark or synthetic datasets rather than validated using diverse real-world organizational data.

# Innovation

- Integrates machine learning-based attrition prediction with Explainable AI.
- Provides understandable explanations for individual employee risk predictions.
- Identifies important employee attributes contributing to the predicted outcome.
- Supports aggregate analysis to identify departments with high attrition rates.
- Can reveal recurring causes, workforce trends and areas requiring management attention.

# Creativity & Novelty

- Creativity: Converts complex machine learning decisions into information that can be understood by HR personnel.

- Novelty: Combines individual prediction, prediction confidence and explainability within one framework.

- Uses multiple possible explanation approaches such as SHAP, LIME, feature importance and permutation importance.
- Extends the analysis from individual employees to department-level workforce patterns.
- Focuses on producing actionable insights rather than only a binary prediction.

# Feasibility

- Technical feasibility: The project uses established supervised machine learning and Explainable AI techniques.

- Data feasibility: The IBM HR Analytics Employee Attrition dataset is identified as a benchmark dataset in the project.
- Implementation feasibility: Data preprocessing, model training, evaluation and visualization can be implemented using Python-based tools.
- Evaluation feasibility: Accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix can be used for model evaluation.
- Project scope is suitable for an academic implementation and can be extended with a dashboard or interface.

# Project Plan

- Phase 1 — Literature survey & requirements
- Phase 2 — Dataset collection & preprocessing
- Phase 3 — EDA & feature engineering
- Phase 4 — ML model development
- Phase 5 — Explainable AI integration
- Phase 6 — Model evaluation
- Phase 7 — Interface/dashboard development
- Phase 8 — Testing, documentation & presentation

# ThankYOU